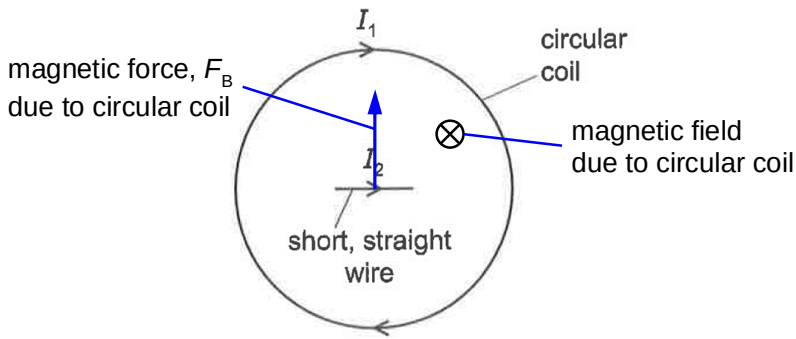


		to the e.m.f. of the battery connected to the circuit.
23	C	In order for the filament lamp to glow more brightly, the p.d. across the lamp has to increase by increasing the resistance of the thermistor. Hence, the thermistor temperature has to decrease. The resistance of the LDR has to decrease, so that the proportion of the p.d. in the circuit is largest across the thermistor. Hence, the light intensity incident on LDR has to increase.

No.	Answer	Solution
24	A	The magnetic field in the circular coil is into the page, by right-hand grip rule. By Fleming's left-hand rule, the magnetic force acting on the short, straight wire due to the circular coil is upwards.  The diagram illustrates a circular coil with current I_1 flowing clockwise. A short, straight wire with current I_2 flows upwards through the center of the coil. A magnetic force F_B acts upwards on the wire. The magnetic field due to the coil is into the page, indicated by a cross in a circle.
25	D	Magnetic flux, $\phi = BA$ Hence, the unit for magnetic flux is tesla metre².
26	A	Magnetic flux linkage in the coil at position X $= NBA$ Magnetic flux linkage in the coil at position Y $= 0$ By Faraday's law, average e.m.f. induced in the coil during the rotation $\langle E \rangle = \frac{\Delta \Phi}{\Delta t}$ $= \frac{3000 \times 1.8 \times \frac{\pi}{4} \times 0.020^2}{0.060}$ $= 28.3$ $\approx 28 \text{ V}$
27	B	Use $\frac{V_s}{V_p} = \frac{I_p}{I_s}$, where $\frac{V_s}{V_p} = \text{gradient of graph} = \frac{17}{2.0}$, Secondary current $I_s = \frac{V_p}{V_s} \times I_p$ $= \frac{2.0}{17} \times 0.32$ $= 0.0376 \text{ A}$ $= 38 \text{ mA}$

No.	Answer	Solution
28	B	Transition from $n = 2$ to $n = 1$, $E_2 - E_1 = \frac{hc}{\lambda_1} \text{ ---- (1) (where } \lambda_1 = 440 \text{ nm)}$ Transition from $n = 3$ to $n = 2$ produces electromagnetic radiation of longer wavelength, i.e. 590 nm. $E_3 - E_2 = \frac{hc}{\lambda_2} \text{ ---- (2) (where } \lambda_2 = 590 \text{ nm)}$ Third transition is transition from $n = 3$ to $n = 1$, $E_3 - E_1 = \frac{hc}{\lambda_3} \text{ ---- (3)}$ (1) + (2) = (3), $\frac{hc}{\lambda_1} + \frac{hc}{\lambda_2} = \frac{hc}{\lambda_3}$ $\frac{1}{\lambda_1} + \frac{1}{\lambda_2} = \frac{1}{\lambda_3}$ $\frac{1}{440} + \frac{1}{590} = \frac{1}{\lambda_3}$ $\lambda_3 = \left(\frac{1}{440} + \frac{1}{590} \right)^{-1}$ $= 252$ $\approx 250 \text{ nm}$
29	D	Intensity $I = \frac{P}{A} = \frac{\frac{E}{t}}{A} = \frac{\frac{N}{t} hf}{A}$ Since frequency remains the same, greater intensity implies greater $\frac{N}{t}$, which is the number of photons passing through unit cross-sectional area of the beam per unit time.
30	B	Mass defect, $\Delta m = 136.90709u - (136.90583u + 5.49 \times 10^{-4}u)$ $= 0.000711u$ By Einstein's mass-energy relationship, $\Delta E = \Delta mc^2$ $= (0.000711 \times 1.66 \times 10^{-27})(3.00 \times 10^8)^2$ $= 1.062 \times 10^{-13}$ $\approx 1.1 \times 10^{-13} \text{ J}$